Nexus Corp Employee Performance and Salary Analysis

Case study: HR & Talent Insights
Role: People Analytics Consultant
Company: Nexus Corp

1. Introduction

Nexus Corp wants to review employee salary and performance information to support fair pay discussions and talent recognition. This report analyzes salary, department, experience, and performance scores for the available employee dataset.

2. Problem Statement

The company needs a simple view of its overall salary benchmark, salary differences between departments, high-performing employees, and the relationship between experience and salary. It also needs to compare average performance across departments.

3. Objectives

1. Calculate the company-wide average monthly salary.
2. Compare average monthly salary by department.
3. Identify employees with high performance scores.
4. Count high performers.
5. examine the relationship between experience and salary.
6. Compare average performance scores by department.

4. Dataset Description

The dataset contains **20 employees** and five original fields:

Field	Description
Employee_ID	Unique employee number

Department	HR, IT, Finance, or Marketing
Years_of_Experience	Employee experience in years
Monthly_Salary	Employee monthly salary
Performance_Score	Performance score from 1 to 10

A new field, **High_Performer**, was created for this analysis. It is **True** when `Performance_Score ≥ 7`; otherwise, it is **False**.

5. Tasks and Results

Task 1: Salary Benchmarking

The average monthly salary across Nexus Corp is:

Measure	Result
Company average monthly salary	**$55,159.65**

This value is the overall compensation benchmark for comparing departments.

Task 2: Departmental Salary Comparison

Department	Employees	Average Monthly Salary
Marketing	6	**$65,270.33**
IT	4	$55,305.50
HR	5	$52,361.40
Finance	5	$45,708.40

Result: Marketing has the highest average monthly salary at **$65,270.33**. Finance has the lowest average monthly salary at **$45,708.40**.

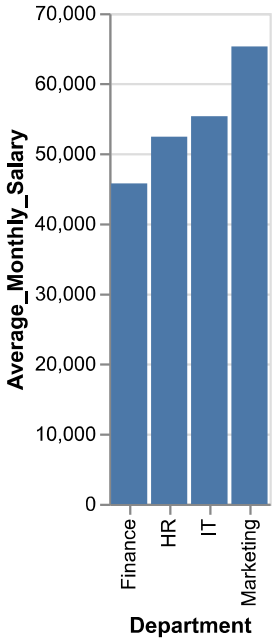


Chart explanation: The chart compares average monthly salary across departments. Marketing is clearly the highest-paying department in this dataset.

Task 3: Identify Top Talent

The following rule was used:

```
df['High_Performer'] = df['Performance_Score'] >= 7
```

Employees with a score of 7 or above were flagged as high performers.

Employee ID	Department	Experience (Years)	Monthly Salary	Performance Score
2010	Finance	1	$34,797	10
2013	Marketing	3	$61,403	10
2018	HR	1	$46,491	9
2007	IT	6	$67,529	8
2005	Finance	12	$55,211	7

2011 Marketing	1	$74,341	7
2014 HR	14	$50,043	7
2017 Marketing	14	$54,695	7

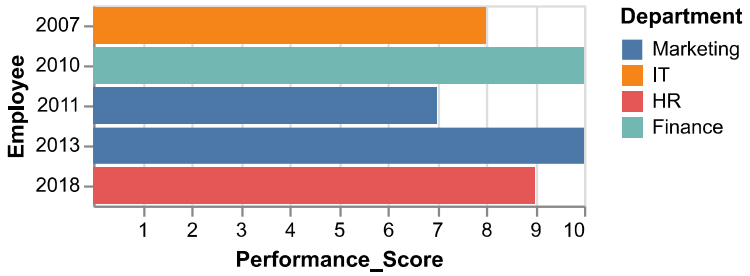


Chart explanation: The chart displays the five highest-scoring employees. Employees 2010 and 2013 achieved the highest score of 10.

Task 4: Headcount Analysis

There are **8 high performers** out of 20 employees, equal to **40%** of the dataset.

Department	High Performers
Marketing	3
Finance	2
HR	2
IT	1

Marketing has the largest number of high performers in this small sample.

Task 5: Experience vs. Salary

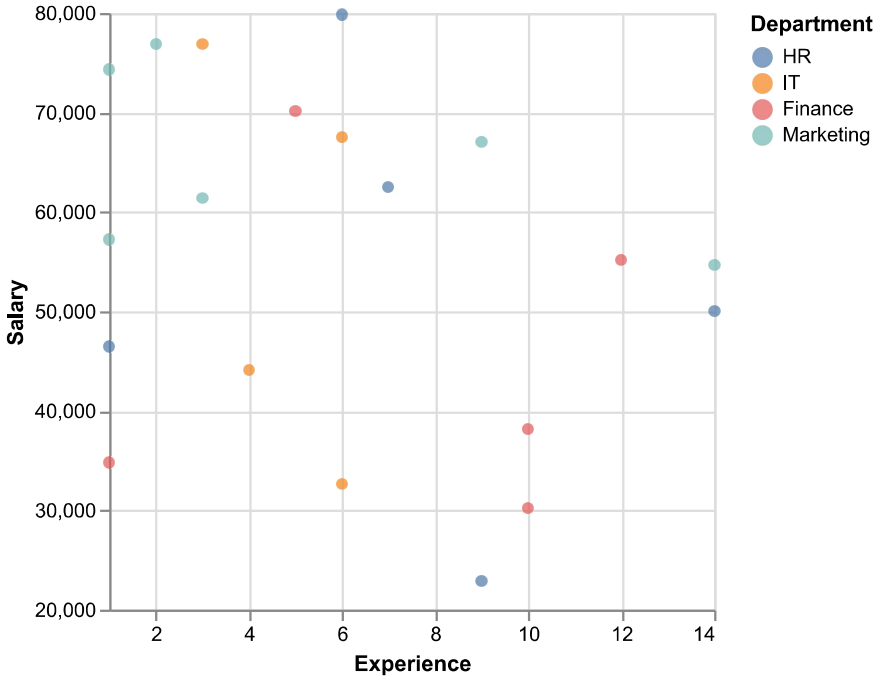


Chart explanation: Each point represents one employee. The chart compares years of experience with monthly salary and uses color for department.

Result: Salary does not show a clear increase as experience rises in this dataset. The Pearson correlation between experience and monthly salary is **-0.277**, which is a weak negative relationship. This result should not be treated as proof that experience lowers salary because the sample contains only 20 employees and does not include job role, level, location, or other pay factors.

Task 6: Performance Review by Department

Department	Average Performance Score
Marketing	**5.83**
Finance	5.80
IT	4.75
HR	4.20

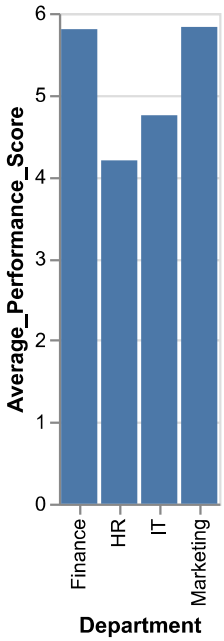


Chart explanation: The chart compares average performance scores by department. Marketing has the highest average score, closely followed by Finance. HR has the lowest average score.

6. Key Findings

- The company-wide average monthly salary is **$55,159.65**.
- **Marketing** has the highest average monthly salary (**$65,270.33**) and the highest average performance score (**5.83**).
- **Finance** has the lowest average salary (**$45,708.40**) but a strong average performance score (**5.80**).
- **8 of 20 employees (40%)** are high performers with scores of 7 or above.
- Experience alone does not appear to explain salary in this dataset.

7. Conclusion

This analysis provides a basic salary and performance overview for Nexus Corp. Marketing stands out for both high average salary and high average performance. Finance may need further review because it has the lowest average salary while maintaining a high average performance score. The results are useful for an initial HR discussion, but they are not a full pay-equity study. A stronger review would need additional data such as job role, grade, location, tenure, education, and demographic information.